Information Assurance and Security – ACIT4630

Iman Anooshehpour
Week 12

BRITISH COLUMBIA
INSTITUTE TECHNOLOGY

BCIT™

Instructional Assessments Online (IAO)

1. Please navigate to this link -> https://my-bcit.bluera.com
2. You'll see "Active Tasks" on the right. You can then select your course CRN from that list of "Active Tasks."
3. Begin your survey.

Learning Objectives

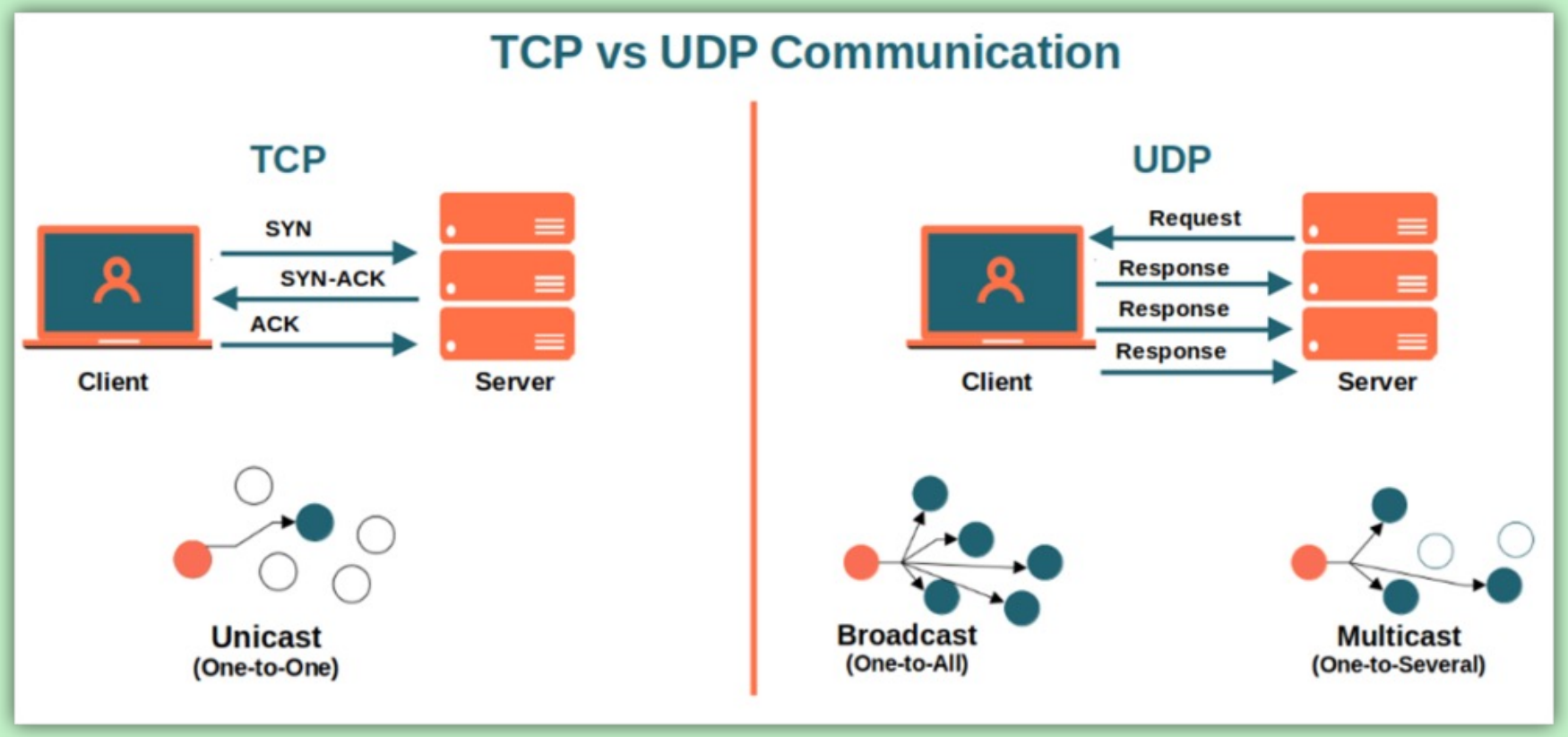
- ❑ List the seven layers in the OSI model
- ❑ List characteristics of TCP protocol and compare it with UDP
- ❑ Calculate the correct seq and ack number in a TCP connection
- ❑ Explain the purpose of packet sniffing and spoofing
- ❑ Perform packet sniffing and spoofing and TCP SYN flood in the lab

OSI Model Open Systems Interconnection

Layer	Name	Role	Protocols	PDU	Address
7	Application	Initiate contact with the network	HTTP, FTP, SMTP	Data	
6	Presentation	Formats data, optional compression, and encapsulation		Data	
5	Session	Initiates, maintains, and tear down session		Data	
4	Transport	Transports data	TCP, UDP	Segment	Port
3	Network	Addressing and routing	IP, ICMP, ARP	Packet	IP
2	Data Link	Frame formation	Ethernet II	Frame	MAC
1	Physical	Data is transmitted on the media		Bits	

OSI Model - Frames

TCP vs UDP



Given the differences between TCP and UDP, in what scenario might choosing UDP be more advantageous than using TCP?

Source

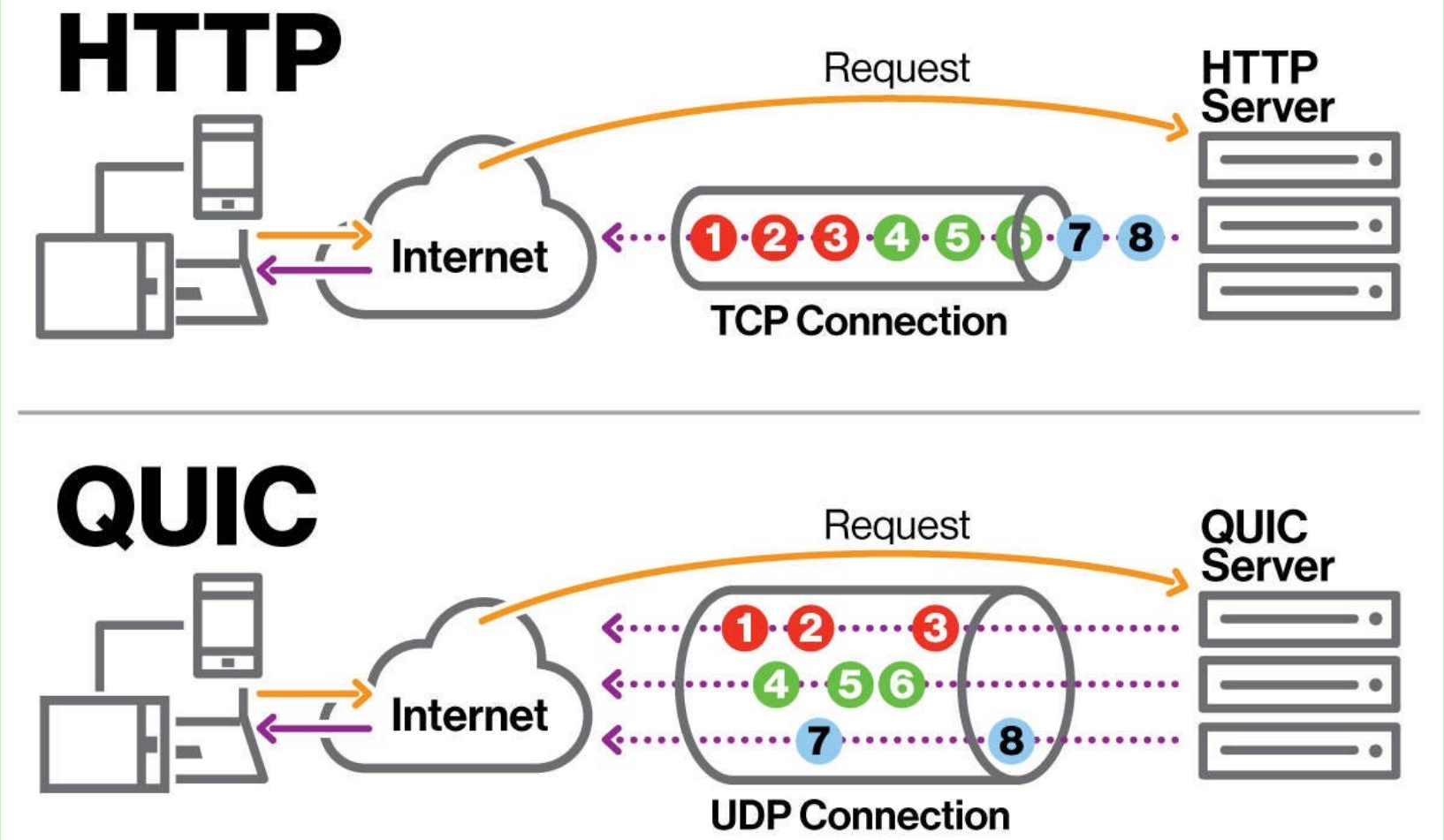
TCP:

- **Reliability:** *TCP provides reliable data delivery by using acknowledgements and retransmissions to ensure all packets reach the destination correctly.*
- **Connection-oriented:** *TCP establishes a connection between the sender and receiver before data transfer, ensuring a reliable and ordered communication stream.*
- **Packet Ordering:** *TCP guarantees that packets arrive at the destination in the same order they were sent, preventing data corruption or loss.*
- **Error Checking:** *TCP performs error checking using checksums to detect and retransmit corrupted packets, ensuring data integrity.*
- **Flow Control:** *TCP utilizes flow control mechanisms to manage the data transmission rate, preventing overwhelming the receiver with a large amount of data.*

UDP:

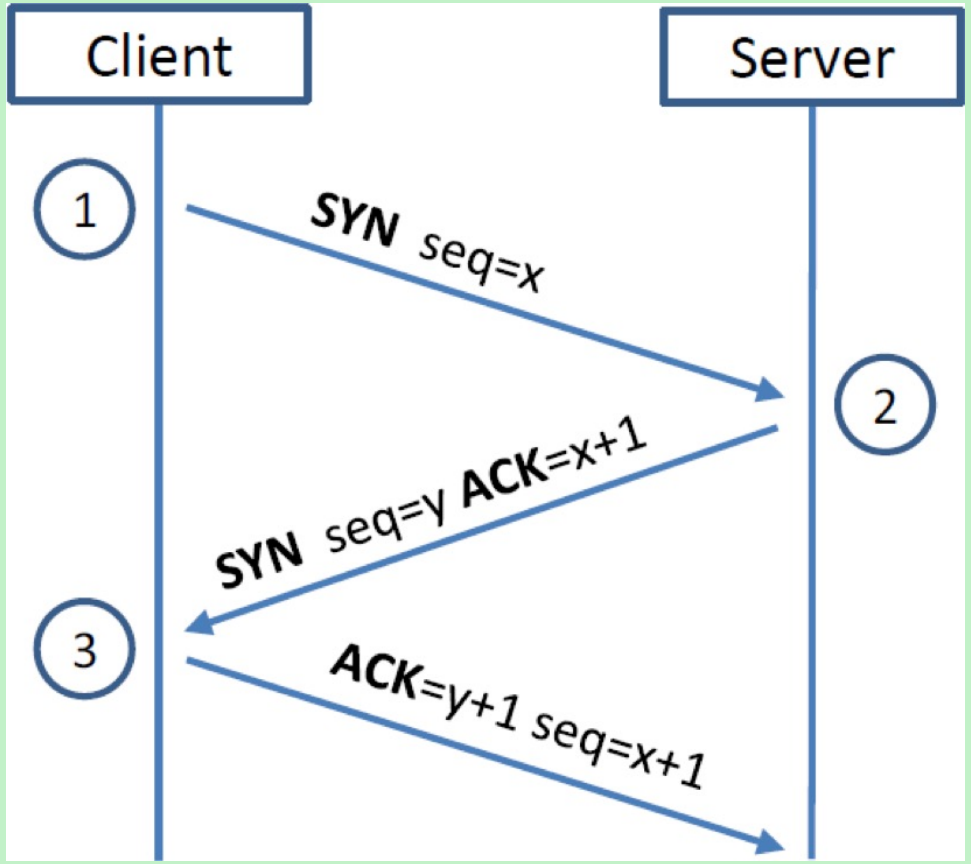
- **Unreliability:** *UDP does not guarantee reliable data delivery or error recovery. It assumes a best-effort delivery approach.*
- **Connectionless:** *UDP does not establish a connection before sending data, making it faster and more lightweight than TCP.*
- **Packet Ordering:** *UDP does not guarantee packet order, which means packets may arrive at the destination out of order.*
- **Error Checking:** *UDP does not perform extensive error checking or retransmission, leaving error handling to higher-level protocols or applications.*
- **No Flow Control:** *UDP does not have built-in flow control mechanisms, allowing for faster transmission but with a higher risk of network congestion*

QUIC (Quick UDP Internet Connections)

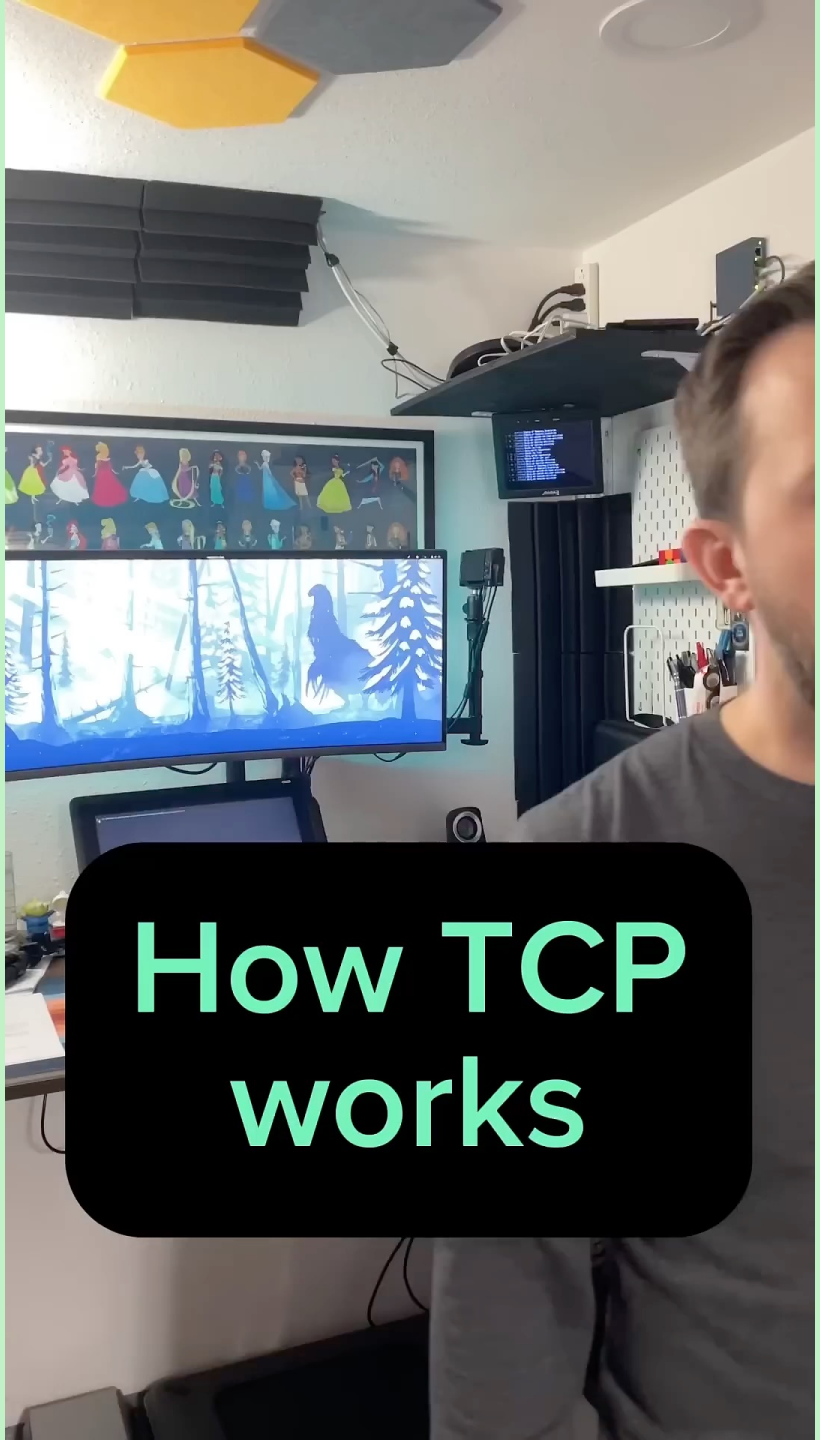


TCP Handshake

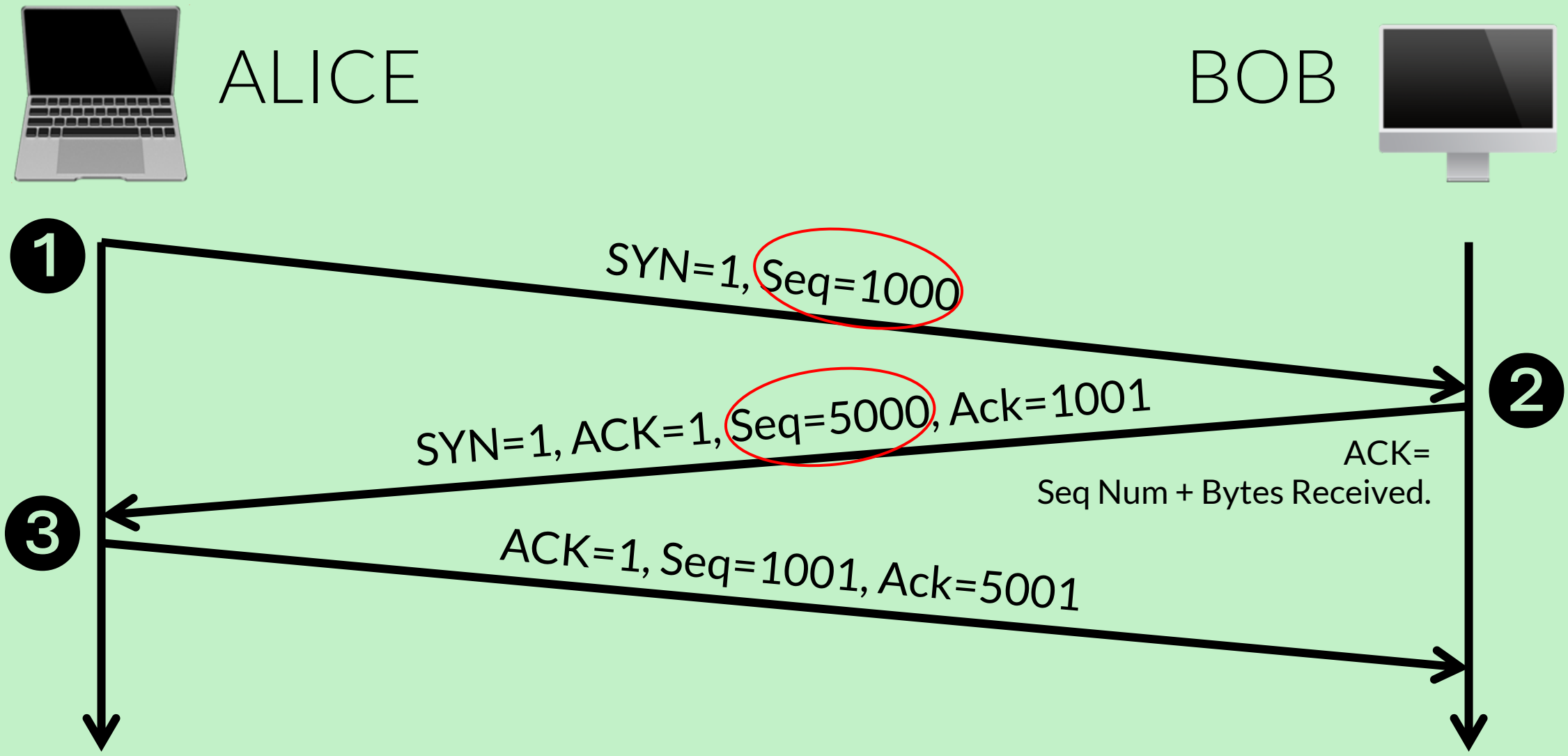
- Why is a SYN packet sent by the server at point 2?
- What is the purpose of following flags in a TCP packet?
  - SYN, ACK, FIN, RST
- Was TCP developed with built-in security mechanism?



TCP IRL



Source

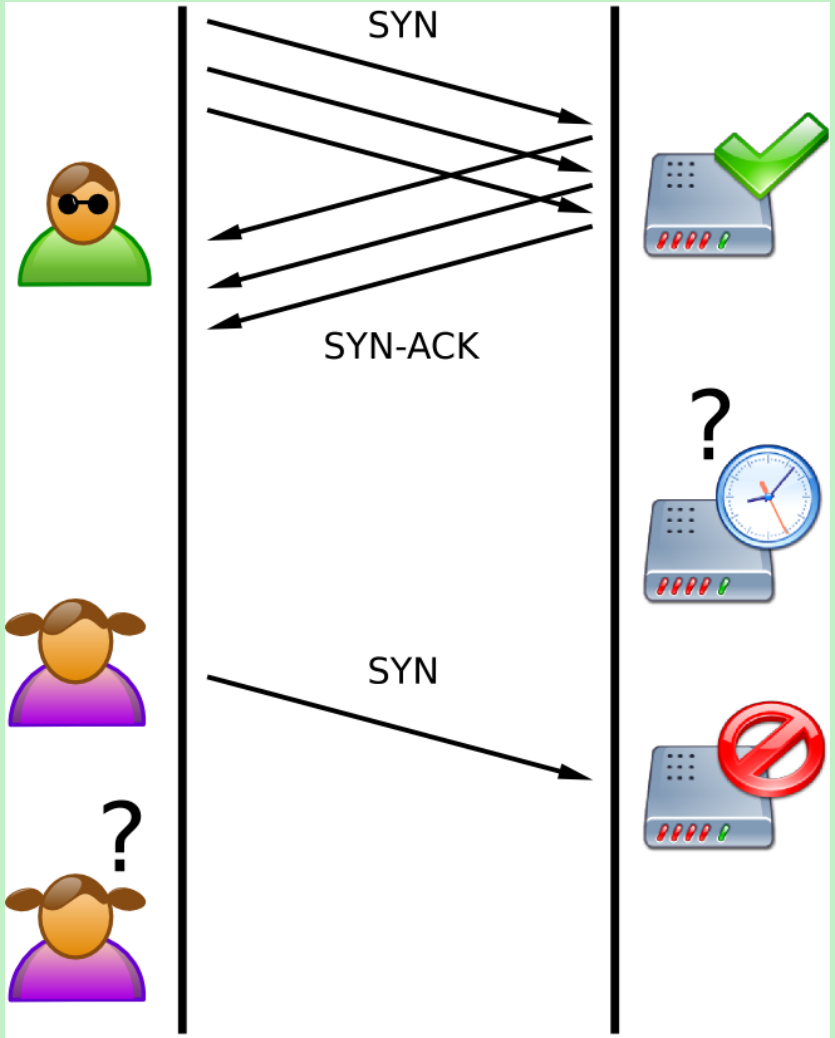


ISN: Initial Sequence Number

```
Alice (Client)                                    Bob (Server)
----------------                                    ----------------

   SYN
   Seq=1000  ---------------------------------------->

                                          SYN + ACK
                                     Seq=5000, ACK=1001
                          <----------------------------------------

   ACK
   Seq=1001, ACK=5001  ---------------------------------------->


=======================  DATA TRANSFER  =========================


   "Hello Bob" (10 bytes)
   Seq=1001, ACK=5001  ---------------------------------------->

                                          ACK
                                     Seq=5001, ACK=1011
                          <----------------------------------------


                                          "Hello Alice" (11 bytes)
                                     Seq=5001, ACK=1011
                          <----------------------------------------

   ACK
   Seq=1011, ACK=5012  ---------------------------------------->


====================  END OF DATA EXCHANGE  ====================
```

SYN Flood Attack

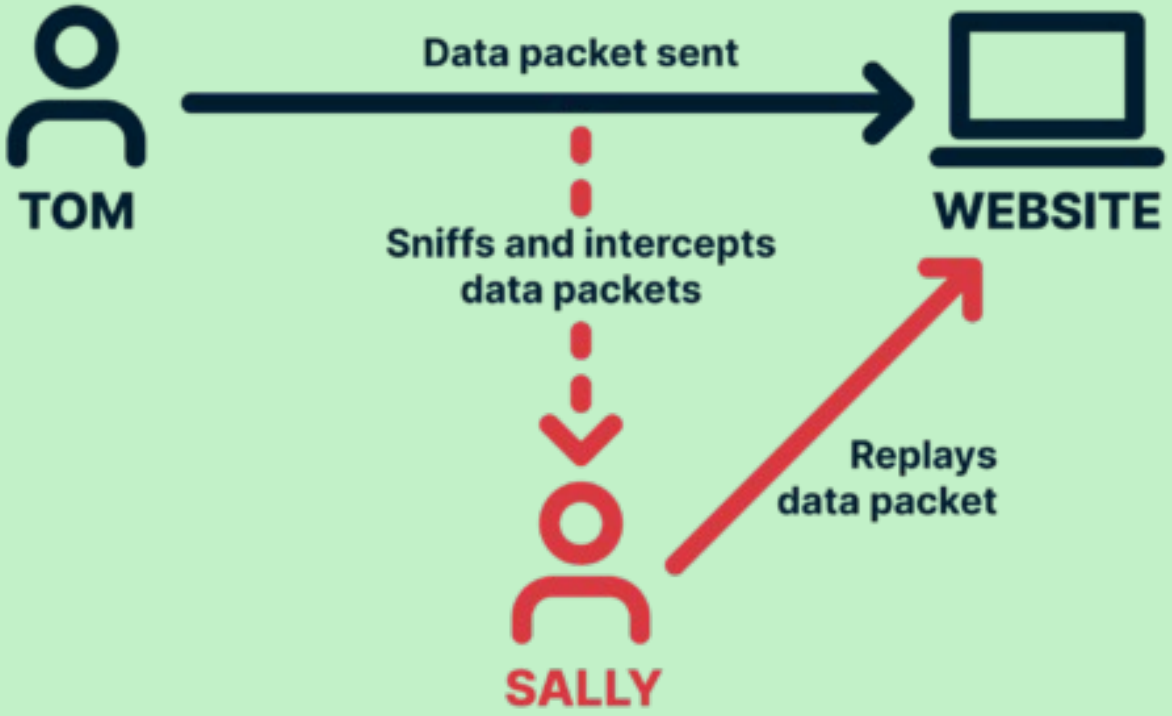
- Which element in the CIA triad is jeopardized in this attack?
- Why is IP spoofing used when attackers send many SYN packets?
- Why are SYN+ACK packets sent to random IPs?
- What happens to the SYN+ACK packets?

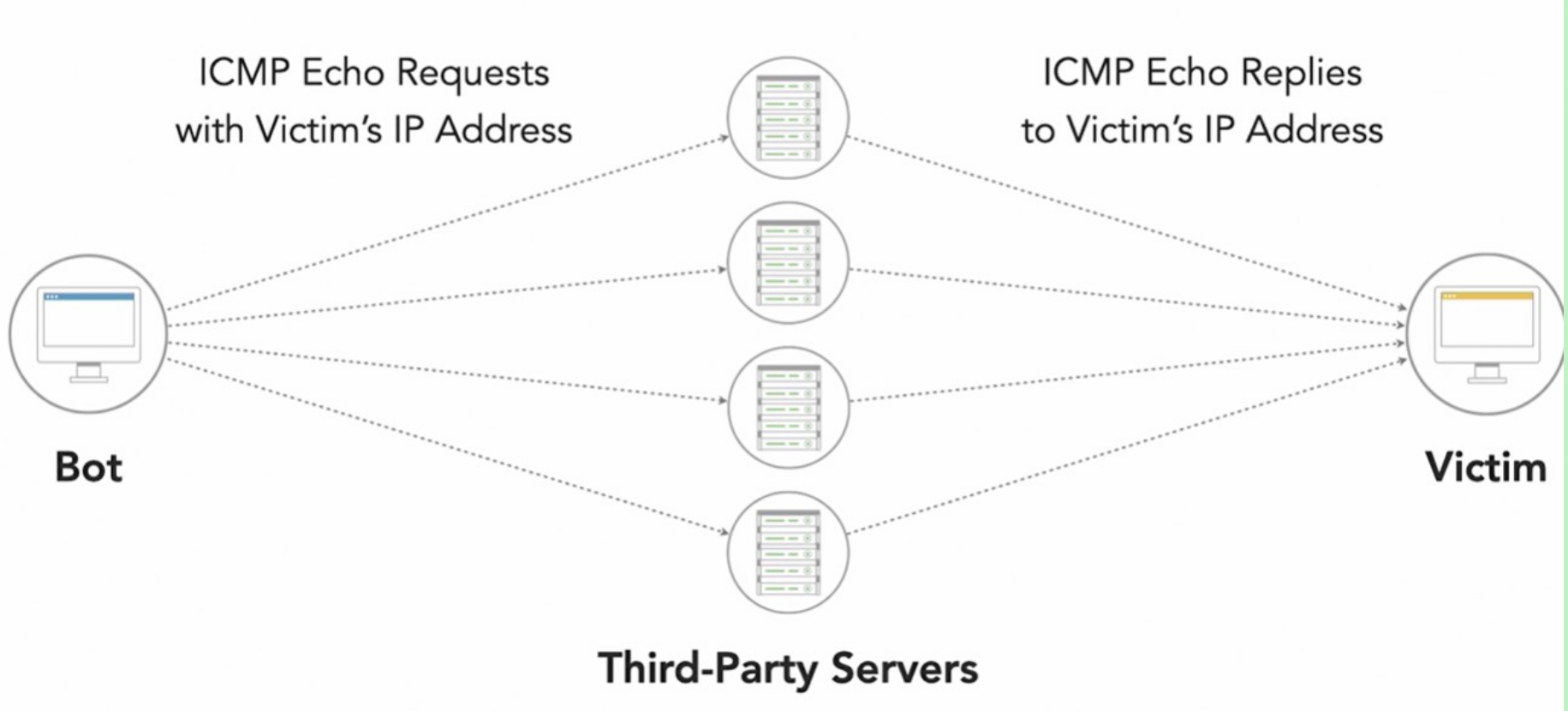


Transmission Control Block queue.

Sniffing (Eavesdropping)

- What is the main goal of Sniffing?
- The attackers rely on a compromised communication path.
  - Name a few ways that this can be achieved.
- What is a replay attack?
  - How to prevent it?



Spoofing

- What is the main goal of Spoofing?
  - How is spoofing used in Smurf attack (an amplified DDoS attack)?